

Radiographic findings and clinical factors in dogs with surgically confirmed or presumed colonic torsion

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Abstract

Colonic torsion is a life-threatening condition in dogs and radiographic findings for this condition have not been well described. The purpose of this retrospective case series was to describe radiographic findings and clinical signs in a group of dogs with colonic torsion. Inclusion criteria were dogs presenting during the period of 2006 and 2016, and that had abdominal radiography and a surgically confirmed or presumed diagnosis of colonic torsion. For each dog, clinical data were recorded from medical records and imaging findings were recorded from retrieved plain radiographs and positive contrast radiographs in which barium enema was performed. Fourteen dogs met inclusion criteria. Of these, nine dogs had colonic torsion confirmed at surgery, with five dogs having surgical confirmation of colonic congestion or mesenteric torsion. Radiographic findings included segmental distention of the colon (14/14), focal narrowing of the colon (11/14), displacement of cecum (11/14), displacement of descending colon (14/14), and mild to no small intestinal distention (14/14). In cases where barium enema was performed, focal narrowing of the colon and longitudinal striations that course in a helical pattern were identified, termed the "torsion sign." Vomiting was the most common clinical sign observed (12/14), followed by abdominal pain in a small majority of cases (8/14). Severe abdominal pain and hypovolemic shock were uncommon in the patients reported (3/14). Colonic torsion should be considered as a differential diagnosis for dogs with radiographic segmental colonic distention with displacement of the descending colon and cecum. Barium enema is recommended for more definitive diagnosis.

KEYWORDS

canine, radiography, small animal, torsion sign

1 | INTRODUCTION

Intestinal torsion and volvulus are life-threatening conditions that can cause ischemia and necrosis of intestinal segments leading to sepsis, circulatory shock, and death.^{1–3} Rapid identification is vital in order to initiate surgical intervention. Intestinal torsion is the pathologic twisting of bowel along its longitudinal axis, while intestinal volvulus is the twisting of bowel around its mesenteric pedicle.^{1,2} The twisting that occurs in either torsion or volvulus can result in partial or complete luminal obstruction depending on the degree of malpositioning. Progressive twisting results in venous and lymphatic compression, vascular engorgement, mural edema of the intestine, ischemia, and ultimately necrosis of the affected intestinal segments.³ Mural ischemia and hypoxia of the intestine results in loss of the mucosal barrier and bacterial endotoxins can then diffuse through the damaged mucosal wall into the systemic circulation, resulting in endotoxemia, circulatory shock, and death.³

Compared to dogs, intestinal volvulus occurs more commonly in humans, horses, cattle, and swine.^{2,4} Risk factors associated with intestinal torsion or volvulus in humans include anatomical variations resulting in increased mobility of the gastrointestinal tract, prior surgery, high fiber intake, chronic constipation, colonic obstruction, hyperperistalsis, and colonic distention.⁵ Prior reports and case series of colonic torsion and volvulus have been described in a total of 17 dogs, occurring primarily in young to middle-aged, medium, and large breed dogs.^{2,6–11} Preexisting or concurrent gastrointestinal disease in these reports include gastric dilatation-volvulus, prior resection and anastomosis of small intestine for foreign body removal, exocrine pancreatic insufficiency, and lymphoplasmacytic enteritis.^{3,4,11} Additionally, a retrospective study found entrapment and strangulation around gastropexy sites in three dogs as a cause of colonic torsion.⁹

Based on the literature, it is unclear if torsion or volvulus of the colon occurred separately or in conjunction with each other in the

cases reported, and these terms have been used interchangeably.^{2,6-11} Given the radiographic and surgical findings we present, the authors will refer to colonic torsion. As colonic torsion, and particularly the imaging findings of this condition, has been infrequently described in dogs, lack of familiarity with this condition and its associated radiographic findings can lead to delays in diagnosis and treatment. Although there have been sporadic case reports of colonic torsion in dogs, the radiographic findings and frequency of those findings have not been well described. Therefore, the purpose of the current study was to describe the radiographic findings associated with colonic torsion in dogs with surgical confirmation and additional cases with presumed colonic torsion. Positive contrast radiographic findings in patients in which barium enema was performed are also described.

2 | METHODS

The study was a retrospective, case series design. A review of the imaging database of the North Carolina State University College of Veterinary Medicine was performed to identify canine patients with a differential diagnosis of colonic torsion on abdominal radiography or keyword of colonic torsion from the period of 2006 through 2016. The North Carolina State University College of Veterinary Medicine surgical database was also searched to identify canine patients with colonic torsion, and cross-referenced with the imaging database. Cases in which colonic torsion was a differential diagnosis were also retrieved from a telemedicine database (Idexx Telemedicine Consultants, IDEXX, Clackamas, OR). Patients had to have at least two orthogonal radiographic views of the abdomen for initial inclusion. Medical records were then reviewed from all cases from both searches to identify dogs that underwent exploratory laparotomy; only patients that underwent exploratory laparotomy were included in the final sample. All decisions for subject inclusion and exclusion were made by all authors, which included two board-certified veterinary radiologists and a veterinarian/small animal intern.

All images were retrieved and reviewed by two board-certified veterinary radiologists and a veterinarian/small animal intern to reach a consensus. Readers were aware of medical record findings at the time of imaging interpretation. Radiographs were evaluated for the following criteria: displacement of the cecum, displacement of the ascending colon, displacement of the transverse colon, displacement of the descending colon, segmental or uniform distension of the colon, colonic fecal content, twisting or narrowing of the colon, and presence of small intestinal distension. Displacement of the cecum and ascending colon was defined as the cecum or ascending colon located to the left of the vertebral column on the ventrodorsal view. Displacement of transverse colon was defined as displacement greater than 45 degrees to the angle of vertebral column on the ventrodorsal view. Displacement of descending colon was defined as displacement to the right of the vertebral column on the ventrodorsal view. Colonic fecal content was assessed subjectively as small, moderate, or large volume. Small intestinal distension was defined as a ratio of greater than 1.6 of the maximal small intestinal diameter to the height of the body of the fifth lumbar vertebra (L5). Medical records were reviewed for the

following clinical information when available: signalment, history of prior gastrointestinal disease or surgery, presenting clinical signs, duration of clinical signs, radiographic findings, time from presentation to surgery, time from imaging to surgery, intraoperative findings, surgical procedure performed, and outcome.

3 | RESULTS

3.1 | Signalment

A total of 19 dogs met initial criteria and five were excluded due to lack of exploratory laparotomy. Fourteen dogs met the final inclusion criteria, five from the North Carolina State University College of Veterinary Medicine and nine from IDEXX. Ages ranged from 2.5 to 14 years (mean 7 years, median 7.25 years), including 10 neutered males, two intact males, and two spayed females. Breeds included Labrador Retriever (5/14), Great Dane (2/14), German Shepherd (2/14), and one of each the following: Irish Wolfhound, Golden Retriever, Gordon Setter, Hound, and English Toy Spaniel.

3.2 | History

Nine of fourteen cases had history of prior gastrointestinal disease or surgery. A total of five of 14 cases had prior gastropexy. Of these, two of 14 had undergone gastropexy for treatment of gastric dilatation-volvulus prior to presentation and the remaining three cases had undergone prophylactic gastropexy. Three of 14 patients had history of prior surgery for foreign body removal, with one patient having resection and anastomosis of 45 cm (18 in) of small intestine 7 years prior to presentation, another patient having undergone enterotomy for foreign body removal, and an unknown surgical procedure for foreign body removal in the third case. Two of 14 cases had a history of chronic weight loss, one of 14 had a history of chronic diarrhea and weight loss, and one of 14 had a history of presumed inflammatory bowel disease that was responsive to tylosin.

3.3 | Presenting clinical signs

The onset of presenting clinical signs was acute to subacute, with duration of 7 h to 7 days prior to presentation (mean 41.6 h, median 30 h). Vomiting was observed in 12/14 dogs. Information regarding frequency of vomiting was not available. Abdominal pain was reported in eight of 14 cases and was characterized as mild in three of 14 dogs and severe in one of 14. Tenesmus was only observed in four of 14 cases. Acute diarrhea was observed in three of 14 cases. Signs of hypovolemic shock were seen in three of 14 cases, with two of these cases having signs consistent with hypovolemic shock on initial presentation. The third patient with signs of hypovolemic shock had mild clinical signs not compatible with hypovolemic shock on presentation and was hospitalized for supportive care. Approximately 14 h following presentation, this patient collapsed and developed acute signs of hypovolemic shock (pale mucous membranes and hypotension) prior to surgery.

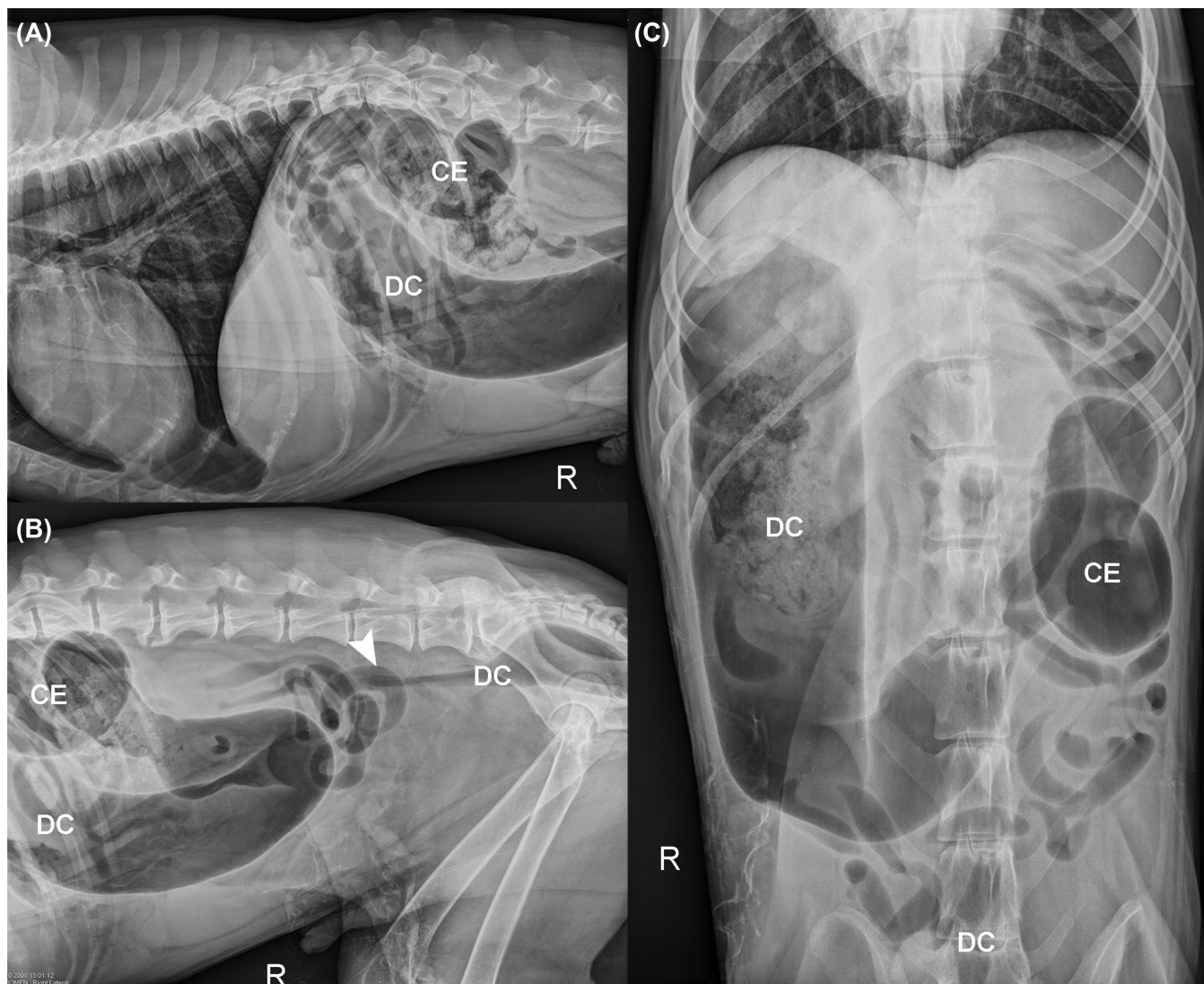


FIGURE 1 Right lateral radiographs of the A, cranial and B, caudal abdomen and a ventrodorsal radiograph (C) of the abdomen of a patient with surgically confirmed colonic torsion. Note the segmental distention of the colon with focal narrowing of the distal descending colon (arrow head). The cecum (CE) is displaced to the left of the vertebral column and the descending colon (DC) is displaced to the right of the vertebral column in the ventrodorsal view

3.4 | Plain radiography

Abdominal radiography was performed in all dogs. Technical parameters used for radiography were not available in the medical records and were not included in the metadata at the time of image reviews. Abdominal radiographic views included right lateral, left lateral, and ventrodorsal projections in 10/14 cases and right lateral and ventrodorsal projections in four of 14 cases. The cecum was identifiable in 11/14 cases, and was displaced in all 11 cases in which it was visible (Figure 1C and Figure 2B). The ascending colon was unable to be clearly identified in 11/14 cases and was displaced in the remaining three cases in which it was visible. The transverse colon was unable to be identified in eight out of 14 cases, but was displaced in the six cases where it could be discerned. The descending colon was displaced in all cases (14/14; Figure 1C and Figure 2B). Segmental distention of the colon was also identified in all cases (14/14; Figure 1A–C and Figure 2A–B). Colonic fecal content was variable, with small volume

(4/14), moderate volume (8/14), and large volume (2/14) noted. Focal narrowing of the descending colon was present in 11/14 cases (Figure 1B and Figure 2A). Small intestinal diameter was normal in the majority of dogs, with two dogs having mild small intestinal distention.

3.5 | Positive contrast radiography

A positive contrast barium enema was performed in four of 14 dogs (Figures 2 and 3). Barium enema was performed in one of these patients using 60% (w/v) barium sulfate suspension (Bracco Diagnostics Inc., Monroe Township, NJ) diluted 1:1 with water administered per rectum at a total dose of 3 ml/kg; radiographs were performed immediately following administration. Other recorded doses included 6, 5, and 10 ml/kg per rectum diluted to 20%. Barium concentration was not recorded for these patients. Focal narrowing of the descending colon with longitudinal striations coursing in a helical pattern was evident in all four cases in which barium enema was performed

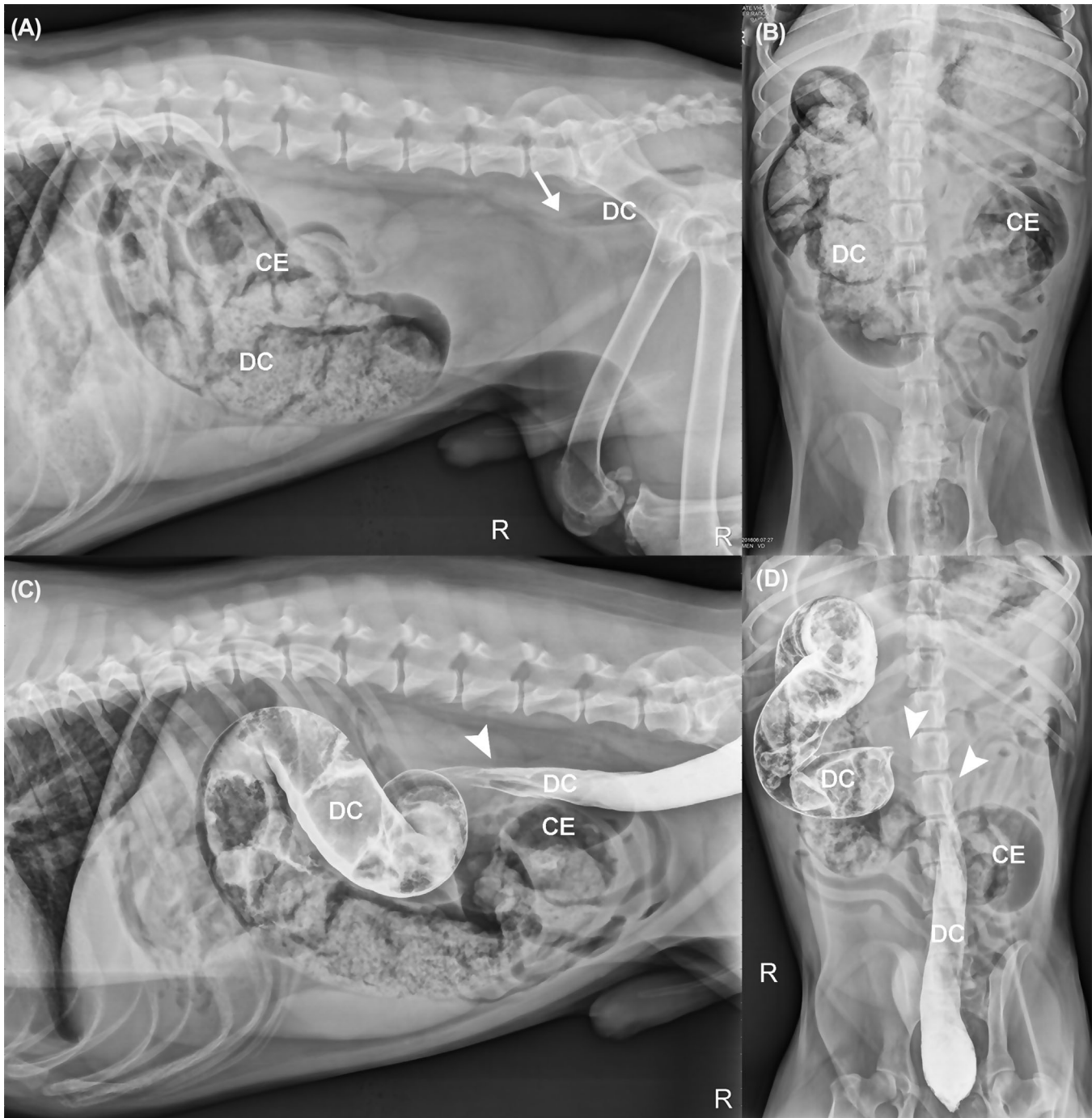


FIGURE 2 Right lateral and ventrodorsal radiographs of the abdomen of a patient with surgically confirmed colonic torsion pre- (A and B) and post-barium enema (C and D). Segmental distention, focal narrowing of the distal descending colon (white arrow), and displacement of the descending colon (DC) and cecum (CE) is seen on pre-contrast images. Post-barium enema, longitudinal, and helical striations are visible in the narrowed section of descending colon (arrow head), termed the “torsion sign”

(Figure 2C and D). In three of these patients, colonic torsion was confirmed at time of surgery. In one of these patients, colonic torsion was not identified at surgery, but the mesentery was hyperemic indicative of prior congestion.

3.6 | Surgical intervention and treatment

Time from presentation to surgery and time from imaging to surgery was available in six of 14 cases. Time from presentation to surgery

ranged from 2 to 20 h (mean 9 h, median 5.5 h). Radiographs at two different time points were performed in two cases. Time from initial radiographic study to surgery ranged from 2 to 18 h (mean 7.6 h, median 4 h). In one of the patients in which radiographs were acquired at two different time points, plain radiography was repeated 15 h after the first study and surgery was performed 3 h following the second set of radiographs. In the second case in which radiographs were acquired at two different time points, a barium enema was performed 10 h after plain radiographs with surgery subsequently performed within 7 h.

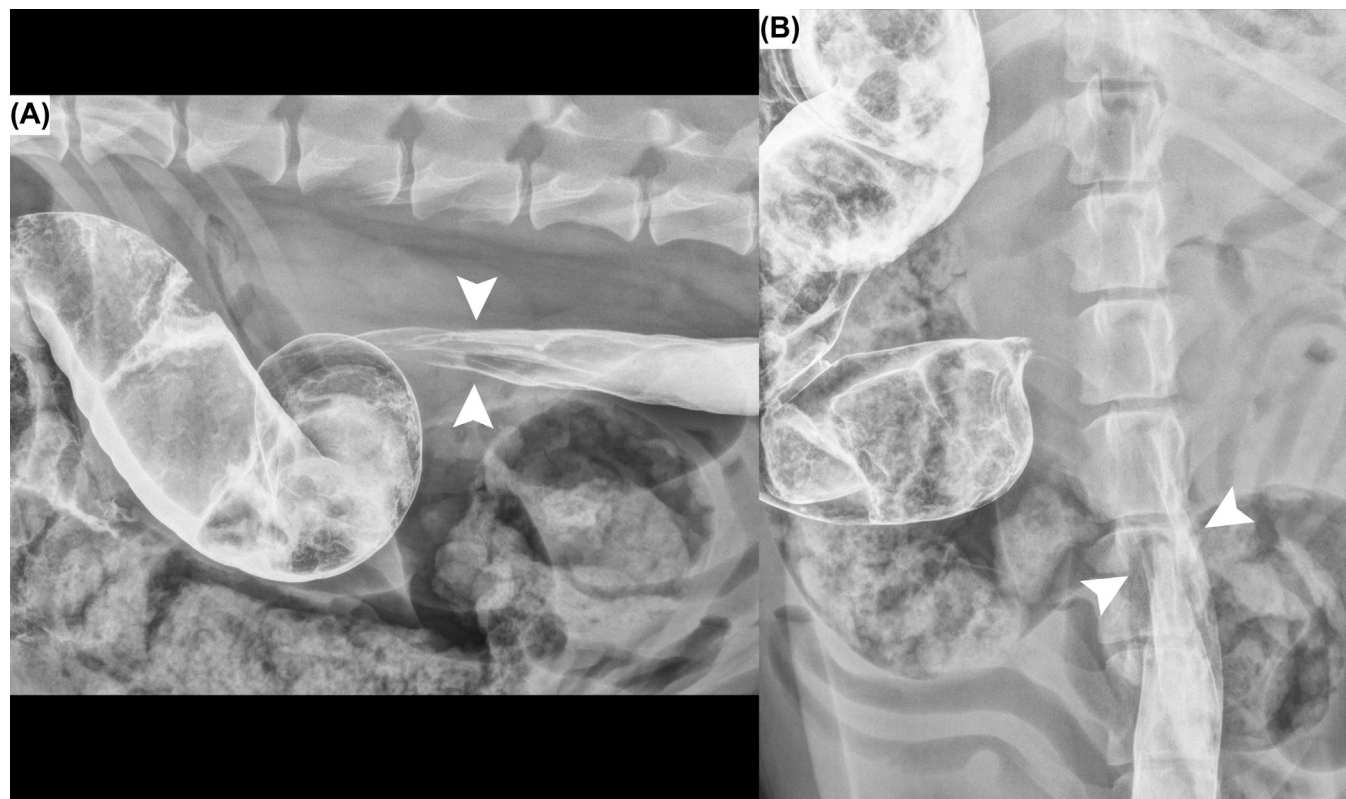


FIGURE 3 Close-up view of right lateral and ventrodorsal radiographs of the abdomen of the same patient in Figure 2 post-barium enema. The longitudinal, helical striations consistent with the “torsion sign” are seen (arrow heads)

Exploratory laparotomy was performed in all dogs. Colonic torsion was confirmed at surgery in nine of 14 cases (64%). Six of nine cases had colonic torsion with a grossly normal, viable colon with no evidence of congestion or necrosis. Two of nine cases had colonic torsion with organ entrapment within their prior gastropexy site, with one case having entrapment of both the torsed colon and spleen in the prior gastropexy site, and the other case having entrapment of spleen in the gastropexy site without involvement of the torsed colon within the gastropexy site. One case had colonic torsion with extensive necrosis of the colon and was euthanized during surgery. One case had gastric foreign material in conjunction with colonic torsion at surgery. Of the eight of nine cases that survived surgery, left colopexy was performed in seven cases. The remaining case had entrapment of the colon in adhesions presumed to be secondary to prior surgery. Manual breakdown of the adhesions was performed in this case without concurrent colopexy. Gastrotomy was performed in the case in which gastric foreign material was identified.

In the five of 14 cases in which colonic torsion was not confirmed at time of surgery, all of the patients had marked gross colonic abnormalities suspicious for prior torsion and four of five of these cases were euthanized or died intra-operatively secondary to their underlying colonic pathology. In one case, the colon was hemorrhagic, congested, and severely gas distended with thrombosis of the vasculature. This patient was euthanized intra-operatively due to the extent of non-viable colon. In another case, the cecum and majority of the colon was hyperemic and markedly gas-distended; subtotal colectomy was discussed with the client intra-operatively and euthanasia was elected.

One patient at surgery had hyperemic and congested colonic mesentery but the colonic wall appeared viable; left colopexy was performed. In an additional patient, the colon was markedly distended and contained black striations and this patient died intra-operatively. The final case had necrosis of the colon in conjunction with mesenteric torsion at the time of surgery and was euthanized intra-operatively due to extensive necrosis of bowel. Necrosis of the colon was considered atypical for mesenteric torsion in this case. This patient was initially stable on presentation and acutely decompensated after radiography, developing signs of hypovolemic shock. Prior to decompensation, abdominal radiographs did not show evidence of small intestinal distention compatible with mesenteric torsion, but rather showed segmental distention of the colon with displacement of the cecum and descending colon consistent with other cases of colonic torsion.

4 | DISCUSSION

Colonic torsion has been previously reported in young to middle-aged dogs with a mean age of 5.2 years (6 months to 10 years) with increased prevalence in male dogs (12/17; 71%) of medium and large breeds (17/17; 100%).^{1,2,6–11} This study supports those findings, as a mean age of 7 years (2.5 to 10 years) was identified, 86% (12/14) were males, and 98% (13/14) were medium to large breed dogs. In prior studies, Great Danes were overrepresented (5/17, 29%)^{2,7,9} compared to this study where Labrador retrievers (5/14; 36%) were overrepresented. However, the overall breed distribution was similar to a prior case

series that included one German shepherd dog, two Great Danes, one Labrador retriever, a Bullmastiff, and a German shorthaired pointer.⁹ To the authors' knowledge this study is the first report of colonic torsion in a small breed dog, an English Toy Spaniel.

In the current study, 64% (9/14) of dogs had history of prior gastrointestinal disease or surgery, with 36% (5/14) of cases having prior gastropexy. In a previous study, three of six dogs had entrapment of their colon around prior gastropexy sites.⁹ This was seen in one case in the current study. Importantly, the majority of patients in this study had mild, nonspecific clinical signs relating to the abdomen and/or gastrointestinal tract. The most common clinical sign noted was vomiting (12/14; 86%). Abdominal pain was present in 57% (8/14) of cases. Of the cases in which abdominal pain was noted, it was graded as severe in only one case. Tenesmus, though intuitively would be expected with colonic torsion, was present infrequently in the current study (4/14; 28% of cases). Similarly, hypovolemic shock was less common in the current study than previously reported.^{1,2,6-11} A prior study reported 83% (5/6) of dogs had signs of hypovolemic shock at presentation, compared to 21% (3/14) in the current study.⁹ The current study highlights the variability and often vague clinical signs in dogs with colonic torsion. Based on these findings, suspicion for colonic torsion should be increased in patients with intermittent clinical signs and/or prior history of gastrointestinal disease, gastric dilatation-volvulus, or prior surgery. Having a high index of suspicion for this condition and identifying pertinent radiographic findings is vital for early surgical intervention in order to decrease the risk of bowel ischemia and necrosis with prolonged torsion.

In this study, 64% (9/14) of cases were confirmed to be colonic torsion at time of surgery. The remaining cases had grossly abnormal colons with evidence of congestion at time of surgery, with one case having necrosis of the colon in conjunction with mesenteric torsion. While speculative, the authors suspect that these cases had dynamic colonic torsion and untorsed between time of imaging to time of surgery. In the case with mesenteric torsion, the patient was initially stable on presentation, but acutely decompensated developing signs of hypovolemic shock following radiography. This patient's abdominal radiographs prior to decompensation showed segmental distention of the colon with displacement of the cecum and descending colon, importantly, without evidence of small intestinal distention. Based on the radiographic findings prior to decompensation, change in the patient's clinical status post-radiography, and necrosis of colon in surgery that is considered atypical in cases of mesenteric torsion, it is speculated that this patient initially had colonic torsion that progressed to mesenteric torsion post-imaging prior to surgery. While this case is speculative given lack of surgical confirmation, it highlights the possibility that colonic torsion may be a dynamic process and the importance of early surgical intervention.

A direct relationship between time from imaging to surgery and clinical outcome could not be made based on the current data, as time from imaging to surgery was only available in six of 14 cases. Mean time from initial radiographic study to surgery was 7.1 h. Two of these cases for which this data was available were euthanized intra-operatively for nonviable colons, with one case taken to surgery within 1 h of radiographs and the other within 5 h of radiographs. Though

surgical intervention was performed sooner than average in these cases, both cases had nonviable colons in surgery and were euthanized intra-operatively. It is possible that surgical intervention was pursued earlier in these cases because their clinical signs were more severe on initial presentation. Two dogs had milder clinical signs on presentation and were hospitalized for supportive care following initial radiographs. Repeat radiographs (plain radiography in one and barium enema in the other) were performed following supportive care, which reinforced the clinical suspicion for colonic torsion and initiated surgical exploration.

Plain radiographic findings in dogs suggestive of colonic torsion in the current study included segmental distention of the colon (14/14), displacement of the descending colon to the right of midline (14/14), focal narrowing of the colon (11/14), displacement of the cecum to the left of midline (11/14), and mild to no small intestinal distention (14/14). In all cases in this study, a definitive diagnosis was not made on plain radiography though colonic torsion was suspected. Barium enema was performed in four of 14 cases. The radiographic characteristics of colonic torsion with barium enema included focal narrowing of the descending colon with longitudinal striations that course in a helical pattern. This appearance occurs secondary to the twisting of the colon around its long axis, resulting in infolding of the colonic wall. Barium coating these mural folds results in helical striations termed the "torsion sign" in this study. The decision to proceed to surgery was aided by barium enema via identification of the "torsion sign" and led to a definitive diagnosis pre-operatively in all four cases.

There are some limitations in this study. This was not a case matched study, so databases were not searched for all cases in which colonic or cecal displacement was present, but a diagnosis of colonic torsion was not made. The method of case selection in this study is a potential source of bias. A surgical database was only available for one institution, so it is possible that there are cases of surgically confirmed colonic torsion which could have radiographic signs different than those reported in this study. As such, other imaging findings could be under-reported. Another possible limitation was the lack of information on the technical parameters that had been used for imaging. However, authors considered all images to be of diagnostic quality. An additional limitation is the lack of a surgical confirmed diagnosis of colonic torsion in all cases in the study. Although five cases did not have colonic torsion confirmed at time of surgery, the radiographic findings were consistent with other cases of colonic torsion, and severe colon pathology (necrosis, congestion, thrombosis) was identified at surgery. As the imaging findings at presentation were suggestive of colonic torsion and their clinical signs matched others, the authors speculate that these cases had colonic torsion at time of imaging that reduced by the time of surgery. While the speculated dynamic component cannot be definitively proven, the authors feel that it is important to report these cases for several reasons. First, given the rarity of colonic torsion as a whole in dogs, the spectrum of this disease may be under-reported if these cases were excluded. Second, these cases support the possibility of colonic torsion as a dynamic process. Third, these cases emphasize the need for high clinical suspicion to prevent delayed time to surgical intervention in these patients, and highlight the importance of early surgical intervention. Lastly, one of the cases developed a marked change in clinical course from having mild

clinical signs and radiographic findings consistent with other cases of colonic torsion, to developing hypovolemic shock with mesenteric torsion and concurrent necrosis of the colon at surgery. This may indicate that disorders of malpositioning can occur concomitantly, both of which in this case would require surgical intervention. Given that 80% (4/5) of the patients without confirmed colonic torsion in this study were euthanized or died at time of surgery and the remaining patient still required a colopexy, surgical intervention was warranted in these patients and was supported by the radiographic findings.

In conclusion, colonic torsion can be a difficult radiographic diagnosis, and clinical signs are often nonspecific and may not be representative of the degree of colonic pathology. The current study describes the radiographic findings of colonic torsion and highlights the utility of barium enema to confirm the diagnosis preoperatively. Many cases of colonic torsion present with mild nonspecific clinical signs, and radiographic diagnosis is key to avoiding delayed surgical treatment. The authors recommend that colonic torsion should be suspected in cases with segmental distention of the colon, focal colonic narrowing, and displacement of the cecum or descending colon. The authors recommend a positive contrast barium enema in cases where colonic torsion is suspected. Future studies are needed to evaluate cases in which colonic or cecal displacement is present alone without concurrent segmental distention of the colon, focal narrowing of the colon, or torsion sign on barium enema and whether or not these patients develop colonic torsion in the future. As some patients have transient colonic or cecal displacement as an apparently incidental finding, further studies would aid in identifying whether these cases represent a spectrum of disease of dynamic torsion or if they are truly incidental.

LIST OF AUTHOR CONTRIBUTIONS

Category 1

- (a) Conception and Design: Gremillion CL, Savage M, Cohen EB
- (b) Acquisition of Data: Gremillion CL, Savage M, Cohen EB
- (c) Analysis and Interpretation of Data: Gremillion CL, Savage M, Cohen EB

Category 2

- (a) Drafting the Article: Gremillion CL, Savage M, Cohen EB
- (b) Revising Article for Intellectual Content: Gremillion CL, Savage M, Cohen EB

Category 3

- (a) Final Approval of the Completed Article: Gremillion CL, Savage M, Cohen EB

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